Airport Management Information System

Organized SQL Management Queries

Course: CMPE343 - Database Management Systems and Programming I

File purpose: This document presents 15 SQL queries prepared for management and statistical reporting. Each query includes a short explanation and a clean SQL block.

Query Overview

Query	Main Purpose
Q1	Retrieve all airplanes with their models and capacity using JOIN.
Q2	Count the number of airplanes per plane model using GROUP BY.
Q3	List technicians who are experts on more than one plane model.
Q4	Find the average test score for each test type.
Q5	Find all airplanes currently parked in a hangar where Out_Date_Time is NULL.
Q6	Get the total number of hours each technician spent on tests, ordered descending.
Q7	Find all traffic controllers whose last medical exam was more than one year ago.
Q8	Extract the month from Test_Date and count tests per month using TO_CHAR.
Q9	List airplanes that scored below 90 in any of their tests.
Q10	Get the full details of the most recent test performed on plane 'TC-JAA'.
Q11	List all employees and output their derived role using outer joins and conditionals.
Q12	Compare total hangar capacity with the current number of stored airplanes using subqueries.
Q13	List airplanes that have not been tested yet using a subquery.
Q14	Find the technician name and test score for the highest score ever recorded.
Q15	Present total maintenance tests and average hours spent per test, grouped by plane model.

Q1. Retrieve all airplanes with their models and capacity using JOIN.

Short explanation: Shows each airplane together with its model manufacturer and capacity by joining Airplane and Plane_Model.

SQL Code:

```
SELECT
    a.Plane_No,
    a.Year_Built,
    m.Manufacturer,
    m.Capacity
FROM Airplane a
JOIN Plane_Model m
    ON a.Model_No = m.Model_No;
```

Q2. Count the number of airplanes per plane model using GROUP BY.

Short explanation: Counts how many airplanes exist for each model number. Useful for fleet distribution.

SQL Code:

```
SELECT
    Model_No,
    COUNT(Plane_No) AS Total_Planes
FROM Airplane
GROUP BY Model_No;
```

Q3. List technicians who are experts on more than one plane model.

Short explanation: Finds technicians who are experts in more than one plane model using GROUP BY and HAVING.

SQL Code:

```
SELECT
    e.Name,
    COUNT(te.Model_No) AS Expertise_Count
FROM Technician_Expertise te
JOIN Employee e
    ON te.SSN = e.SSN
GROUP BY e.Name
HAVING COUNT(te.Model_No) > 1;
```

Q4. Find the average test score for each test type.

Short explanation: Calculates the average score for each maintenance test type.

SQL Code:

```
SELECT
    Test_ID,
    AVG(Score) AS Avg_Score
FROM Maintenance_Test
GROUP BY Test_ID;
```

Q5. Find all airplanes currently parked in a hangar where Out_Date_Time is NULL.

Short explanation: Lists airplanes that are currently in a hangar by checking records without an OUT date/time.

SQL Code:

```
SELECT
    Plane_No,
    Hangar_No,
    In_Date_Time
FROM Airplane_Location
WHERE Out_Date_Time IS NULL;
```

Q6. Get the total number of hours each technician spent on tests, ordered descending.

Short explanation: Ranks technicians by the total time they spent on maintenance tests.

SQL Code:

```
SELECT
    e.Name,
    SUM(mt.Hours_Spent) AS Total_Hours
FROM Maintenance_Test mt
JOIN Employee e
    ON mt.SSN = e.SSN
GROUP BY e.Name
ORDER BY Total_Hours DESC;
```

Q7. Find all traffic controllers whose last medical exam was more than one year ago.

Short explanation: Identifies traffic controllers whose medical examination date is older than one year.

SQL Code:

```
SELECT
    e.Name,
    tc.Last_Exam_Date
FROM Traffic_Controller tc
JOIN Employee e
    ON tc.SSN = e.SSN
WHERE tc.Last_Exam_Date < CURRENT_DATE - INTERVAL '1 year';
```

Q8. Extract the month from Test_Date and count tests per month using TO_CHAR.

Short explanation: Groups test events by month and counts how many tests were performed in each month.

SQL Code:

```
SELECT
    TO_CHAR(Test_Date, 'Month') AS Test_Month,
    COUNT(Event_ID) AS Total_Tests
FROM Maintenance_Test
GROUP BY TO_CHAR(Test_Date, 'Month');
```

Q9. List airplanes that scored below 90 in any of their tests.

Short explanation: Finds airplanes that received a score below 90 in at least one test.

SQL Code:

```
SELECT DISTINCT
    Plane_No
FROM Maintenance_Test
WHERE Score < 90;
```

Q10. Get the full details of the most recent test performed on plane 'TC-JAA'.

Short explanation: Retrieves the most recent test record for a selected airplane.

SQL Code:

```
SELECT
    *
FROM Maintenance_Test
WHERE Plane_No = 'TC-JAA'
ORDER BY Test_Date DESC
LIMIT 1;
```

Q11. List all employees and output their derived role using outer joins and conditionals.

Short explanation: Lists employees and classifies their role as Controller, Technician, or Other using left joins and CASE.

SQL Code:

```
SELECT
    e.Name,
    e.Union_Mem_No,
```

```
    CASE
        WHEN tc.SSN IS NOT NULL THEN 'Controller'
        WHEN t.SSN IS NOT NULL THEN 'Technician'
        ELSE 'Other'
    END AS Emp_Type
FROM Employee e
LEFT JOIN Traffic_Controller tc
    ON e.SSN = tc.SSN
LEFT JOIN Technician t
    ON e.SSN = t.SSN;
```

Q12. Compare total hangar capacity with the current number of stored airplanes using subqueries.

Short explanation: Compares total hangar capacity against the number of airplanes currently stored.

SQL Code:

```
SELECT
    (SELECT SUM(Capacity) FROM Hangar) AS Max_Capacity,
    (SELECT COUNT(*) FROM Airplane_Location WHERE Out_Date_Time IS NULL) AS
        Current_Stored;
```

Q13. List airplanes that have not been tested yet using a subquery.

Short explanation: Finds airplanes that do not have any maintenance test records yet.

SQL Code:

```
SELECT
    Plane_No
FROM Airplane
WHERE Plane_No NOT IN (
    SELECT DISTINCT Plane_No
    FROM Maintenance_Test
);
```

Q14. Find the technician name and test score for the highest score ever recorded.

Short explanation: Shows the technician and score for the highest test score recorded in the system.

SQL Code:

```
SELECT
    e.Name,
    mt.Score
FROM Maintenance_Test mt
JOIN Employee e
    ON mt.SSN = e.SSN
WHERE mt.Score = (
    SELECT MAX(Score)
    FROM Maintenance_Test
);
```

Q15. Present total maintenance tests and average hours spent per test, grouped by plane model.

Short explanation: Summarizes test count and average hours spent per test for each airplane model.

SQL Code:

```
SELECT
    a.Model_No,
    COUNT(mt.Event_ID) AS Total_Tests,
    ROUND(AVG(mt.Hours_Spent), 2) AS Avg_Hours
FROM Maintenance_Test mt
JOIN Airplane a
    ON mt.Plane_No = a.Plane_No
GROUP BY a.Model_No;
```